DESIGN AND DEVELOPMENT OF FREELANCER-CLIENT PLATFORM USING DESIGN THINKING APPROACH

A MINI PROJECT REPORT FOR THE COURSE

GE 23627 DESIGN THINKING AND INNOVATION

Submitted by

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in

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BONAFIDE CERTIFICATE

Certified that the mini project titled **"FREELANCER-CLIENT PLATFORM"** is the bonafide work carried out by **Aditya S (240701020)**, **Rohith Kumar (240701436)** under my supervision. This project has been completed in partial fulfilment of the requirements of the course **GE23621 – Design Thinking and Innovation**, promoting project-based learning and practical application of design principles.

Certified further that, to the best of my knowledge and belief, the work reported herein is original and does not form part of any other project, report, or work submitted previously for any academic award.

This project contributes toward achieving the following **Sustainable Development Goals (SDGs):**

1. Decent Work and Economic Growth (SDG 8)
2. Industry, Innovation and Infrastructure (SDG 9)
3. Reduced Inequalities (SDG 10)

Student Signature with Name

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Signature of the Supervisor with date

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Design and Development of a Freelancer–Client Platform Using Design Thinking

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***Abstract*—Freelancing has become a key component of the modern digital economy, enabling individuals and organizations to collaborate across geographical boundaries. Despite the success of platforms such as Fiverr and Upwork, many challenges still exist, including inefficient communication, lack of structured workflows, poor transparency, and difficulty in tracking project progress.**

This project presents the design and development of a Freelancer–Client platform using the Design Thinking methodology. The system is implemented as a full-stack web application that integrates job posting, proposal management, messaging, file submission, and role-based dashboards.

The platform aims to provide a structured workflow, improve communication, and enhance transparency throughout the project lifecycle. The application of Design Thinking ensures a user-centered approach, resulting in improved usability and efficiency. :contentReference[oaicite:0]index=0

***Index Terms*—Freelancing, Design Thinking, Web Application, Full Stack, Workflow Management**

I. Introduction

Freelancing has experienced rapid growth and has become an essential part of the global workforce. With the advancement of digital technologies, individuals can offer services such as web development, graphic design, and content creation to clients worldwide.

However, existing freelancing platforms face several issues. Freelancers often encounter unclear requirements, delayed feedback, and lack of structured submission systems. Clients struggle with tracking progress, managing multiple freelancers, and verifying deliverables.

The primary issue is the absence of a centralized and structured workflow. This leads to inefficiencies, miscommunication, and reduced productivity. To address these problems, this project proposes a Freelancer–Client platform that integrates all functionalities into a unified system.

A. Objectives

The main objectives of the project are:

- To design a structured workflow for freelance project management
- To improve communication between clients and freelancers
- To provide secure file submission and tracking
- To enhance transparency and accountability
- To develop a scalable full-stack web application

II. Design Thinking Methodology

Design Thinking is a human-centered approach that focuses on understanding user needs and solving problems creatively. It emphasizes empathy, ideation, prototyping, and testing.

A. Empathize

In this stage, user needs are identified through research and observation. Interviews and analysis of existing platforms were conducted to understand common issues.

B. Define

The core problem identified is the lack of structured workflow and integrated communication systems in freelancing platforms.

C. Ideate

Various solutions were proposed, including dashboards, messaging systems, and file tracking mechanisms.

D. Prototype

A working prototype was developed using modern web technologies.

E. Test

The system was tested with users, and feedback was used to improve the platform.

III. System Architecture

The platform follows a client-server architecture.

A. Frontend

The frontend is developed using React, providing a responsive and interactive user interface.

B. Backend

The backend is built using Node.js and Express, handling API requests, authentication, and business logic.

C. Database

MongoDB is used to store user data, job details, and messages.

D. Authentication

JWT (JSON Web Token) is used for secure authentication and authorization.

IV. Modules Description

A. User Authentication

Users can register and log in securely. Role-based access control ensures proper functionality for clients and freelancers.

B. Job Posting

Clients can post jobs with detailed requirements.

C. Proposal System

Freelancers can submit proposals, and clients can approve or reject them.

D. Messaging System

A real-time messaging system enables communication between users.

E. File Submission

Freelancers can upload project files, and clients can review them.

F. Dashboard

Each user has a personalized dashboard to track activities.

V. Implementation

The system is implemented using the MERN stack:

- MongoDB
- Express.js
- React.js
- Node.js

RESTful APIs are used for communication between frontend and backend.

VI. Testing and Results

The system was tested for:

- Functional correctness
- Performance
- Security
- Usability

The results indicate improved workflow efficiency and better communication.

VII. Advantages

- Structured workflow
- Improved communication
- Secure file handling
- Better user experience
- Scalable architecture

VIII. Limitations

- No payment integration
- Limited real-time features
- Basic UI design

IX. Future Work

Future improvements include:

- Payment gateway integration
- Real-time chat using WebSockets
- AI-based recommendations
- Mobile application development
- Analytics dashboard

X. Conclusion

The Freelancer–Client platform successfully provides a structured solution for managing freelance projects. The integration of Design Thinking ensures that the system meets user needs effectively.

The platform improves communication, enhances transparency, and simplifies workflow management. It has the potential to be extended into a fully functional commercial product.

Acknowledgment

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